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CampusGuard is a system used to manage a university's response to certain incidents, such as medical emergencies or fires. When an incident occurs, many parties are involved, such as the security team, the facilities, the medical team, etc., however not all of them need to know of each other at the same time. This coupling problem is what our design is aiming to solve.

[illegible]

Command participants:

- Command – Command
- ConcreteCommand – SecureAreaCommand, EvacuateZoneCommand, DispatchUnitCommand, ActivateAlertServiceCommand, RequestExternalAssistCommand, CancelLastActionCommand
- Invoker – CommandInvoker
- Receiver – DispatchCOLleague, AccessControlSystem, AlertService, CampusGuardService, CommandInvoker
- Client – EmergencyFacade

Mediator participants:

Colleague – Colleague

ConcreteColleague – SecurityColleague, MedicalColleague, FacilitiesColleague, AccessControlColleague, CommunicationColleague, MutualAidColleague

Mediator – not implemented, as we do not have multiple concrete classes so there is no need for an abstract class.

ConcreteMediator - IncidentMediator

Adapter participants:

Target – CampusGuardService

Adaptee – LegacyService

Adapter – LegacyServiceAdapter

Client - EmergencyFacade

Facade participants:

Facade – EmergencyFacade

SubClasses – IncidentRepository, IncidentMediator, CommandInvoker,

AccessControlSystem, AlertService, CampusGuardService

Client - CampusGuardSystem

Our Pattern Choices Justification

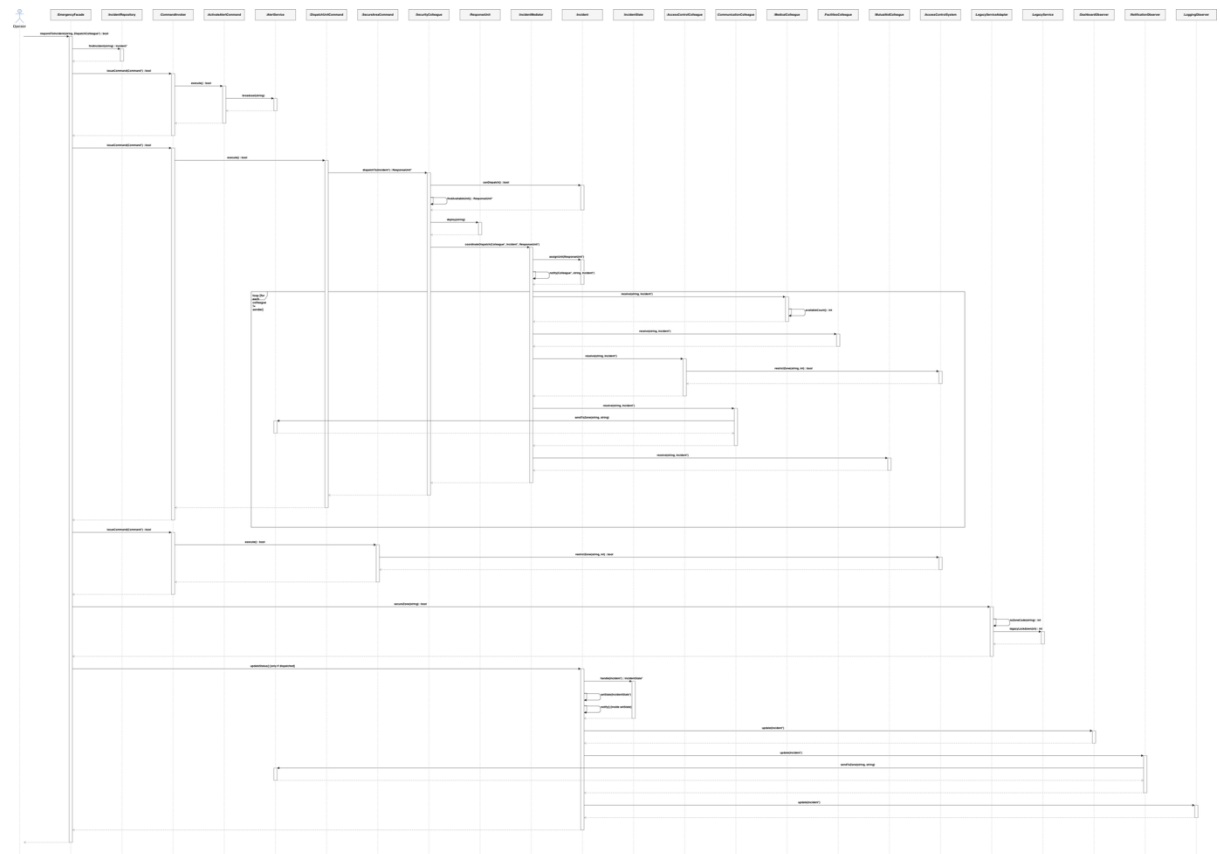
We chose to implement the State and Observer patterns.

State is critical in managing what an incident is allowed to do at different points. For example, our system trying to handle an incident when it is in the Reported state would allow us to dispatch the necessary units, whereas if the incident is already in the Resolved state, no further action is necessary.

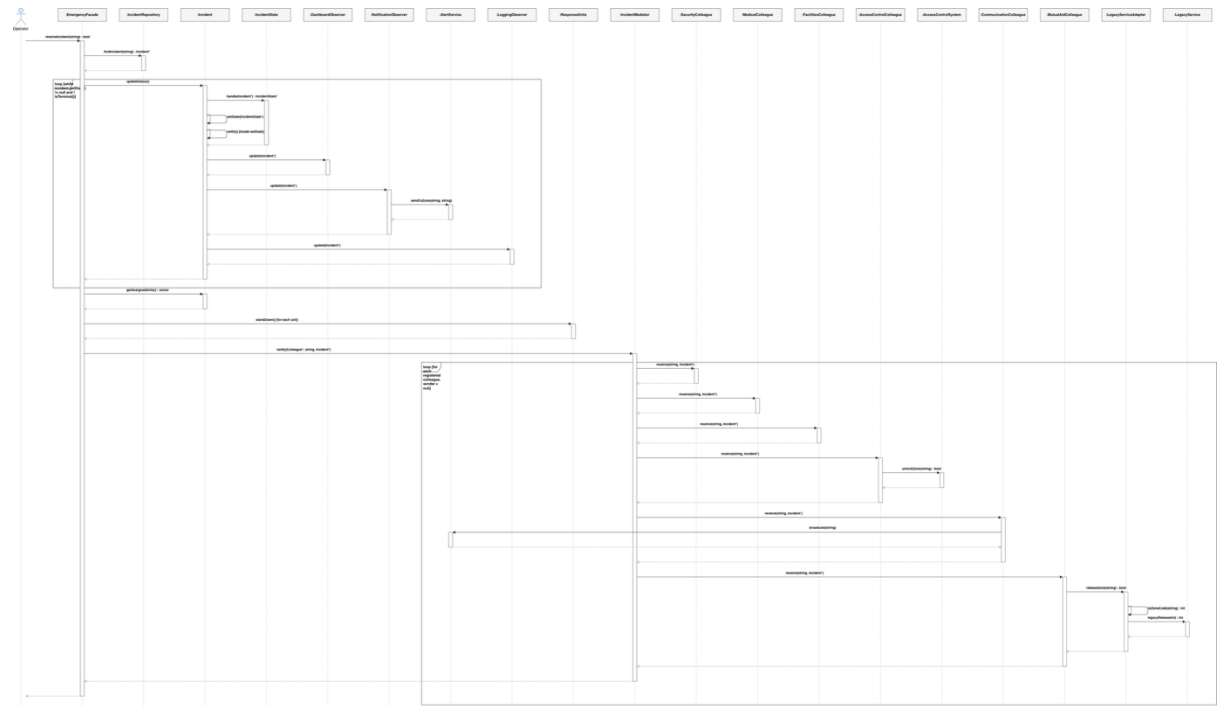
Observer is used to communicate the aforementioned State changes to the necessary participants. Our ConcreteObservers can print the current status, send a message to the AlertService, and log the audit trail (the events that occurred).

Behavioural Diagrams

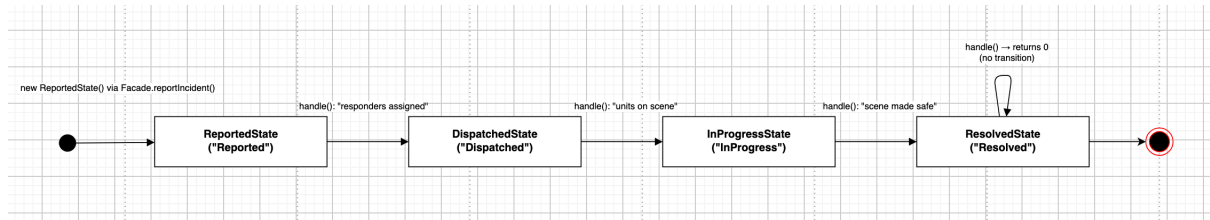
Sequence Diagram 1:



Sequence Diagram 2:



State Diagram:



GDB Investigation and Valgrind Evidence

GDB:

```
1 (gdb) set pagination off
2 (gdb) set logging file gdb_session.txt
3 (gdb) set logging enabled on
4 Copying output to gdb_session.txt.
5 Copying debug output to gdb_session.txt.
6 (gdb) break IncidentMediator::notify
7 Breakpoint 1 at 0x1520a: file IncidentMediator.cpp, line 20.
8 (gdb) run
9 Starting program: /mnt/c/Users/entle/OneDrive - University of Pretoria/Year 2/Semester 2/COS214/Practicals/Practical 5/COS214-PRACT5/campusGuard
10
11 This GDB supports auto-downloading debuginfo from the following URLs:
12 <https://debuginfod.ubuntu.com>
13 Enable debuginfod for this session? (y or [n]) y
14 Debuginfod has been enabled.
15 To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
16 Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
17 Downloading separate debug info for /lib/x86_64-linux-gnu/libstdc++.so.6
18 Downloading separate debug info for /lib/x86_64-linux-gnu/libgcc_s.so.1
19 [Thread debugging using libthread_db enabled]
20 Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
21
22 =====
23 CampusGuard: Emergency Response Coordination
24 =====
25 [Startup] 6 response components registered with the coordinator.
26
27 =====
28 SCENARIO 1: Fire alarm, Engineering Building A
29 =====
30
31 --- Facade::reportIncident ---
32 [Dashboard] INC-001 (Fire, severity 5) @ Engineering Building A -> Reported
33 [AlertService] -> ZONE-A: Incident INC-001 is now Reported
34 [AuditLog] recorded: INC-001 | Reported | Engineering Building A
35 [AlertService] -> ZONE-A: An incident has been reported in your area.
36 [Facade] Zone check: Engineering Building A [Unlocked, level 0]
37
38 --- Facade::respondToIncident ---
39 [Command] ActivateAlertCommand -> Alert Service
40 [AlertService] CAMPUS BROADCAST: Emergency response underway for INC-001 at Engineering Building A
41 [Command] DispatchUnitCommand -> Facilities Desk
42 [Facilities FAC-01] On scene at Engineering Building A.
43 -> utilities isolated, hazard barriers placed.
44 [Mediator] Coordinating Facilities unit FAC-01 on scene at Engineering Building A.
45
46 Breakpoint 1, IncidentMediator::notify (this=0x555555906e0, sender=0x55555590be0, event="unit_dispatched", incident=0x555555910f0) at IncidentMediator.cpp:20
47 20 Incident* incident) {
48 (gdb) print colleagues_.size()
49 $1 = 6
50 (gdb) print colleagues_[0]@8
51 $2 = {0x55555590b00, 0x55555590b70, 0x55555590be0, 0x55555590c50, 0x55555590c00, 0x0, 0x0}
52 (gdb) break IncidentMediator.cpp:27 if i == 6
53 Breakpoint 2 at 0x555555693af: file IncidentMediator.cpp, line 27.
54 (gdb) continue
55 Continuing.
56 [Mediator] 'unit_dispatched' raised by Facilities Desk for INC-001 -> informing 5 component(s).
57 [Security Desk] clearing an approach route for Engineering Building A.
58 [Medical Desk] preparing a casualty point (1 unit(s) still free).
59 [AccessZone ZONE-A] access restricted to level 3 (responders only).
60 [AlertService] -> ZONE-A: Responders are on scene. Follow their instructions.
61
62 Breakpoint 2, IncidentMediator::notify (this=0x555555906e0, sender=0x55555590be0, event="unit_dispatched", incident=0x555555910f0) at IncidentMediator.cpp:27
63 27 if (colleagues_[i] && colleagues_[i] != sender)
64 (gdb) print i
65 $3 = 6
66 (gdb) print colleagues_.size()
67 $4 = 6
68 (gdb) print colleagues_[i]
69 $5 = (Colleague *) 0x0
70 (gdb) list 26,30
71 26 for (size_t i = 0; i <= colleagues_.size(); ++i) {
72 27 if (colleagues_[i] && colleagues_[i] != sender)
73 28 colleagues_[i]->receive(event, incident);
74 29 }
75 30 }
76 (gdb) set logging enabled off
77 Done logging to gdb_session.txt.
```

Valgrind:

```
==1332==
==1332== HEAP SUMMARY:
==1332==    in use at exit: 0 bytes in 0 blocks
==1332==   total heap usage: 204 allocs, 204 frees, 87,793 bytes allocated
==1332==
==1332== All heap blocks were freed -- no leaks are possible
==1332==
==1332== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

Dockerfile and Run Instructions

docker-compose.yml:

```
1  services:
2    campusguard:
3      build: .
4      container_name: campusguard
5      command: ./campusGuard
```

dockerfile:

```
1  FROM gcc:latest
2
3  WORKDIR /app
4
5  RUN apt-get update && apt-get install -y valgrind && rm -rf /var/lib/apt/lists/*
6
7  COPY *.h *.cpp Makefile ./
8
9  RUN make
10
11 CMD [ "campusGuard" ]
```

Github, Reflection, and Statements

<https://github.com/u25622103-droid/COS214-PRAC5>

We all worked as a team and helped each other out in every section of the work. No design decision was finalized until every member approved it. Each member did an equal amount of work

Github helped us to push the most recent version of our design/diagrams, such that the other team members could remain updated and create the code accordingly. It also helped us in the form of Pull Requests, as different team members could work on the

code concurrently and then use a PR to merge them together, allowing us to work more efficiently without sacrificing the integrity of our design.